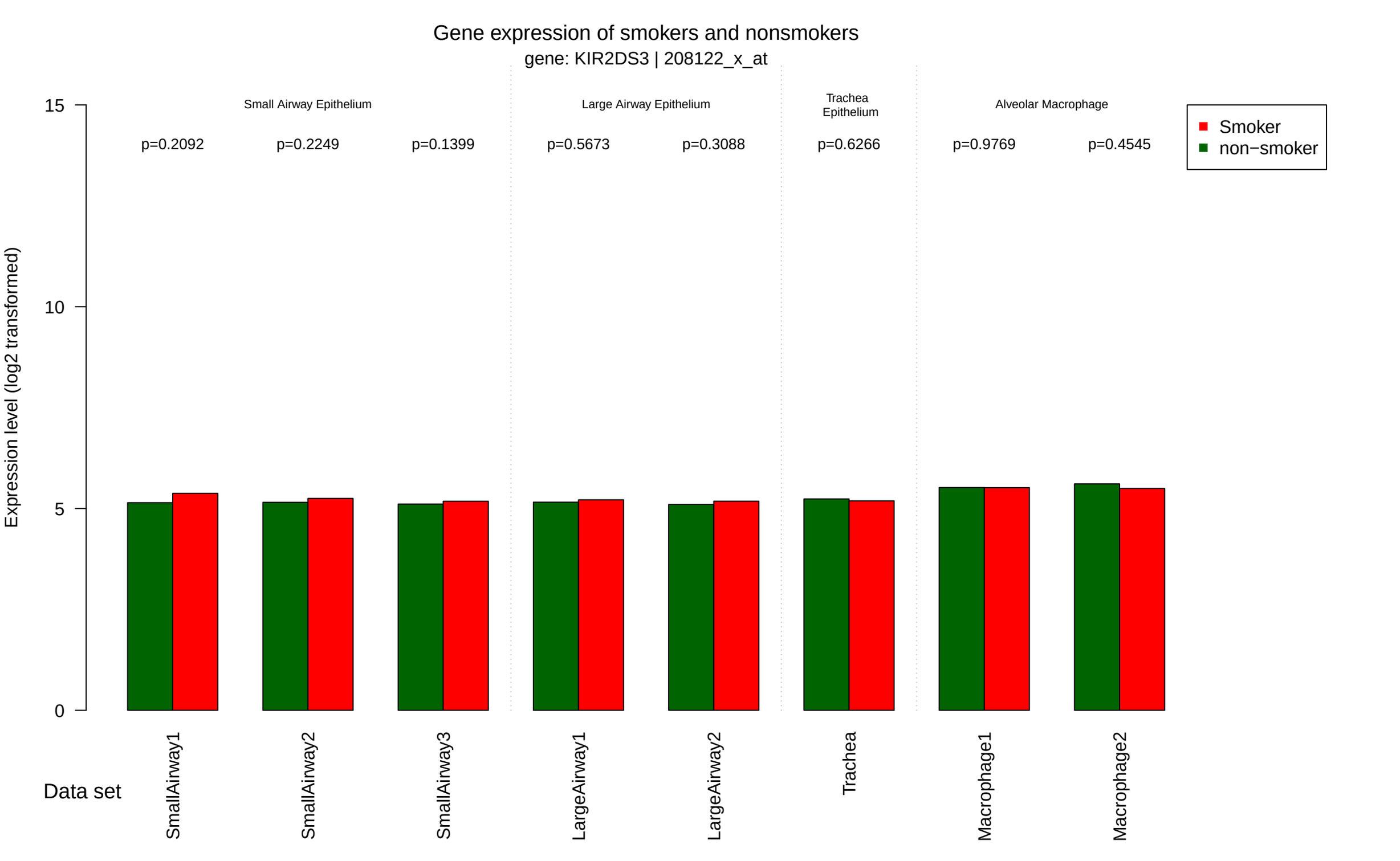
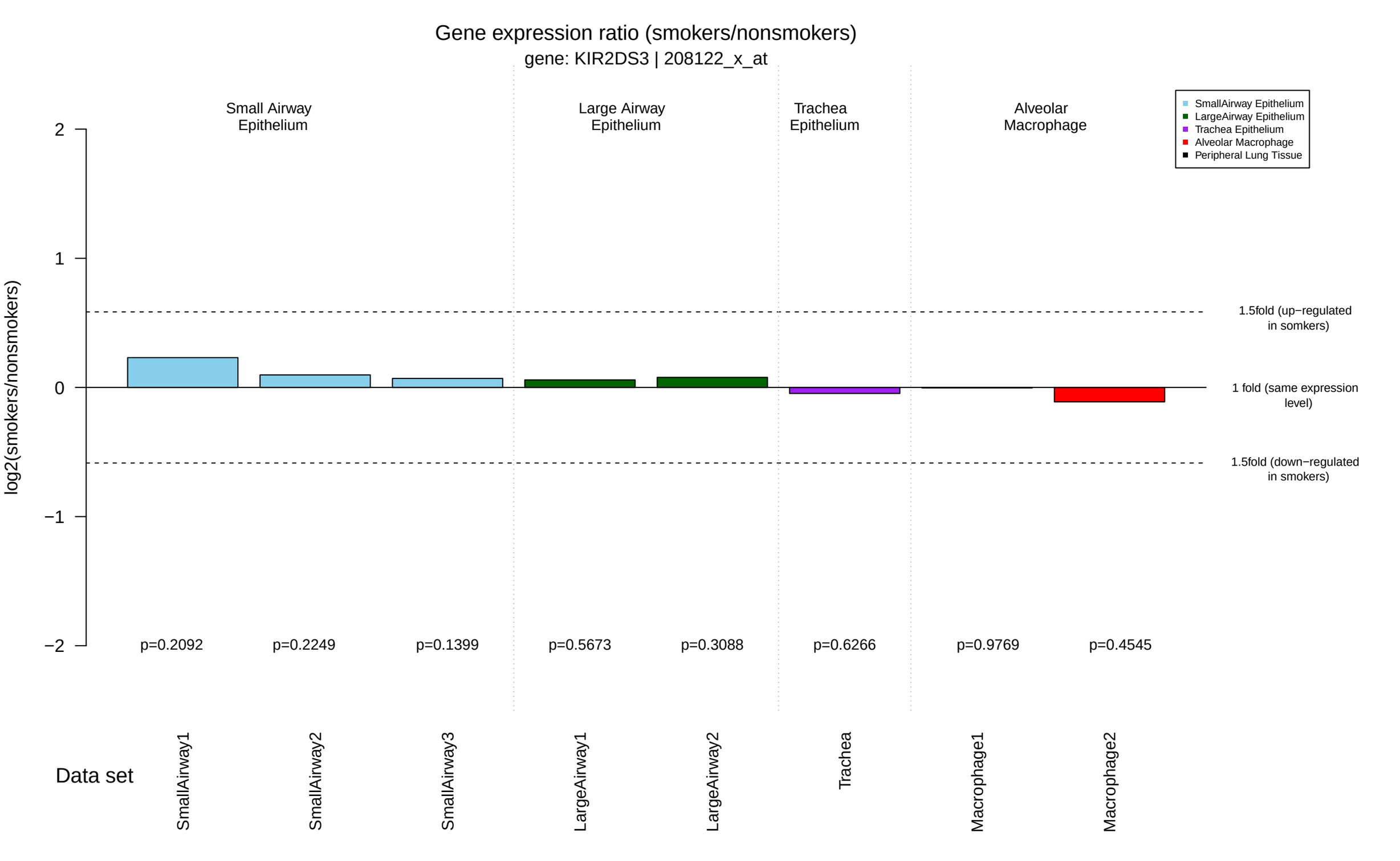
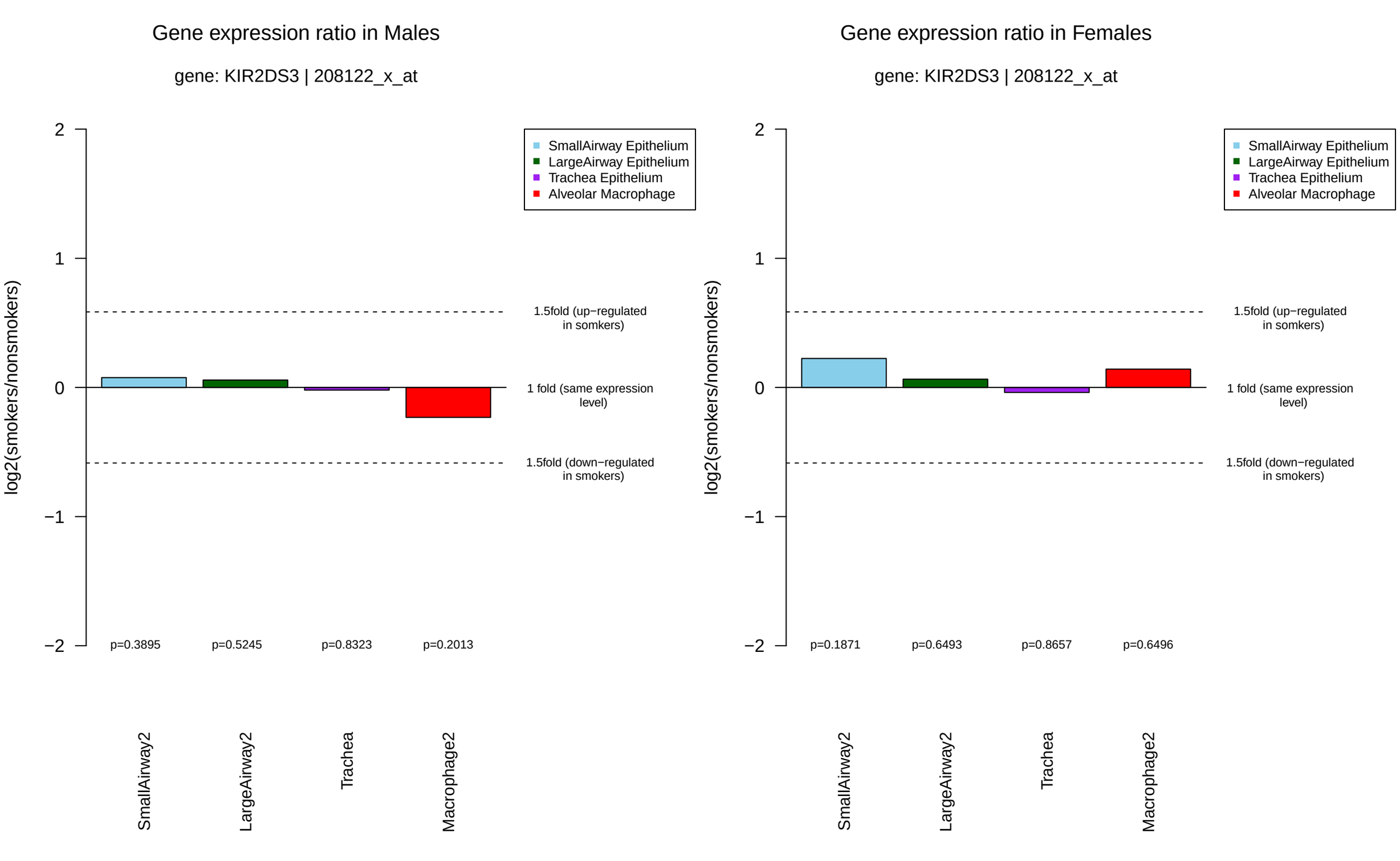
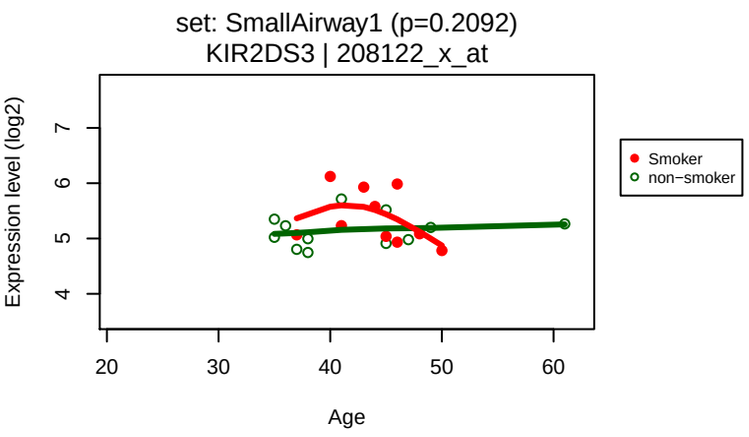
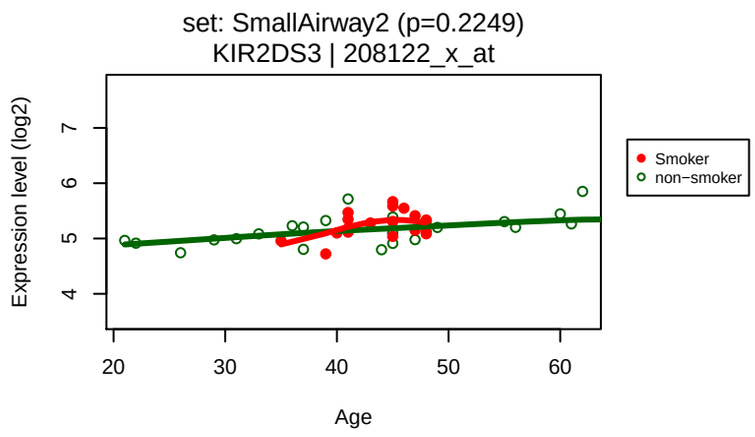
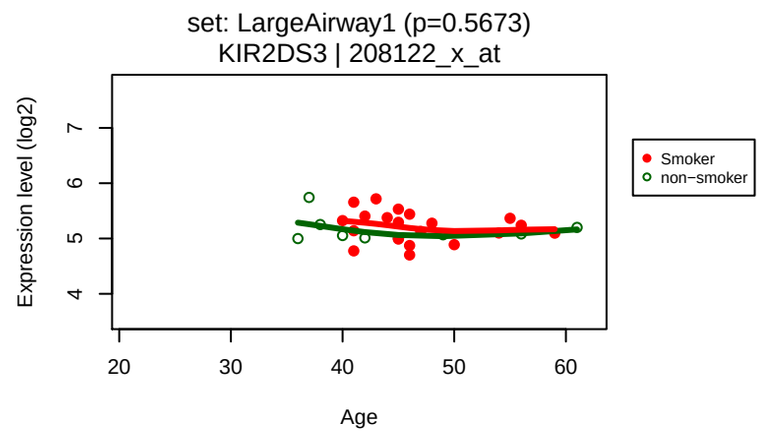

Gene expression of smokers and nonsmokers
gene: KIR2DS3 | 208122_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2092
p=0.2249
p=0.1399
p=0.5673
p=0.3088
p=0.6266
p=0.9769
p=0.4545
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

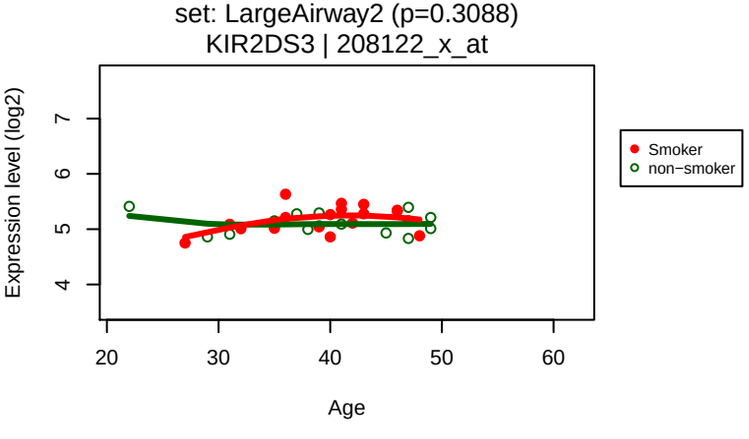

Gene expression ratio (smokers/nonsmokers)
gene: KIR2DS3 | 208122_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2092
p=0.2249
p=0.1399
p=0.5673
p=0.3088
p=0.6266
p=0.9769
p=0.4545
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

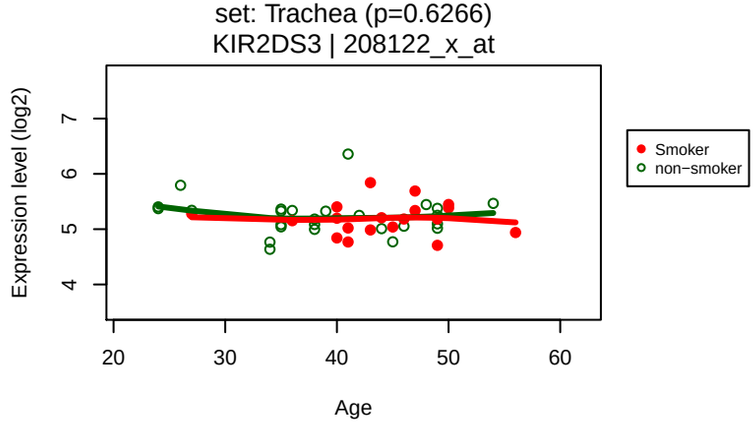

Gene expression ratio in Males
gene: KIR2DS3 | 208122_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3895
p=0.5245
p=0.8323
p=0.2013
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIR2DS3 | 208122_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1871
p=0.6493
p=0.8657
p=0.6496
SmallAirway2
LargeAirway2
Trachea
Macrophage2

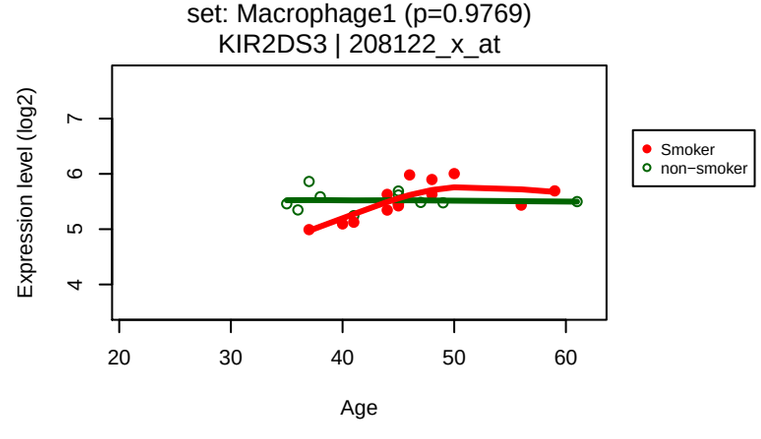

set: SmallAirway1 (p=0.2092)
KIR2DS3 | 208122_x_at
Expression level (log2)
Age
Smoker
non−smoker

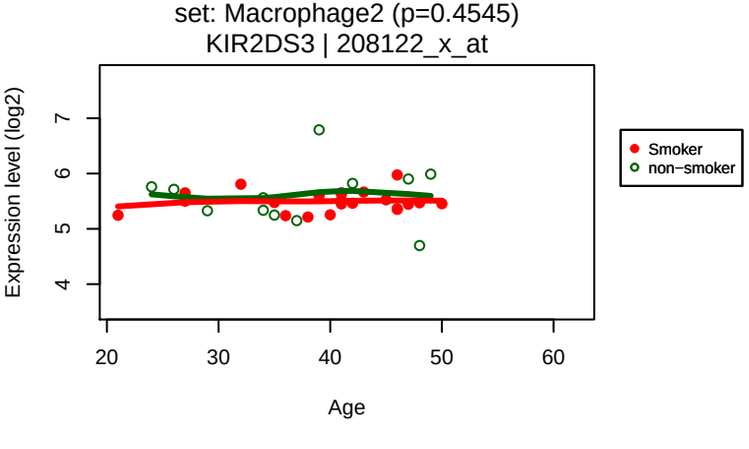

set: SmallAirway2 (p=0.2249)
KIR2DS3 | 208122_x_at
Expression level (log2)
Age
Smoker
non−smoker

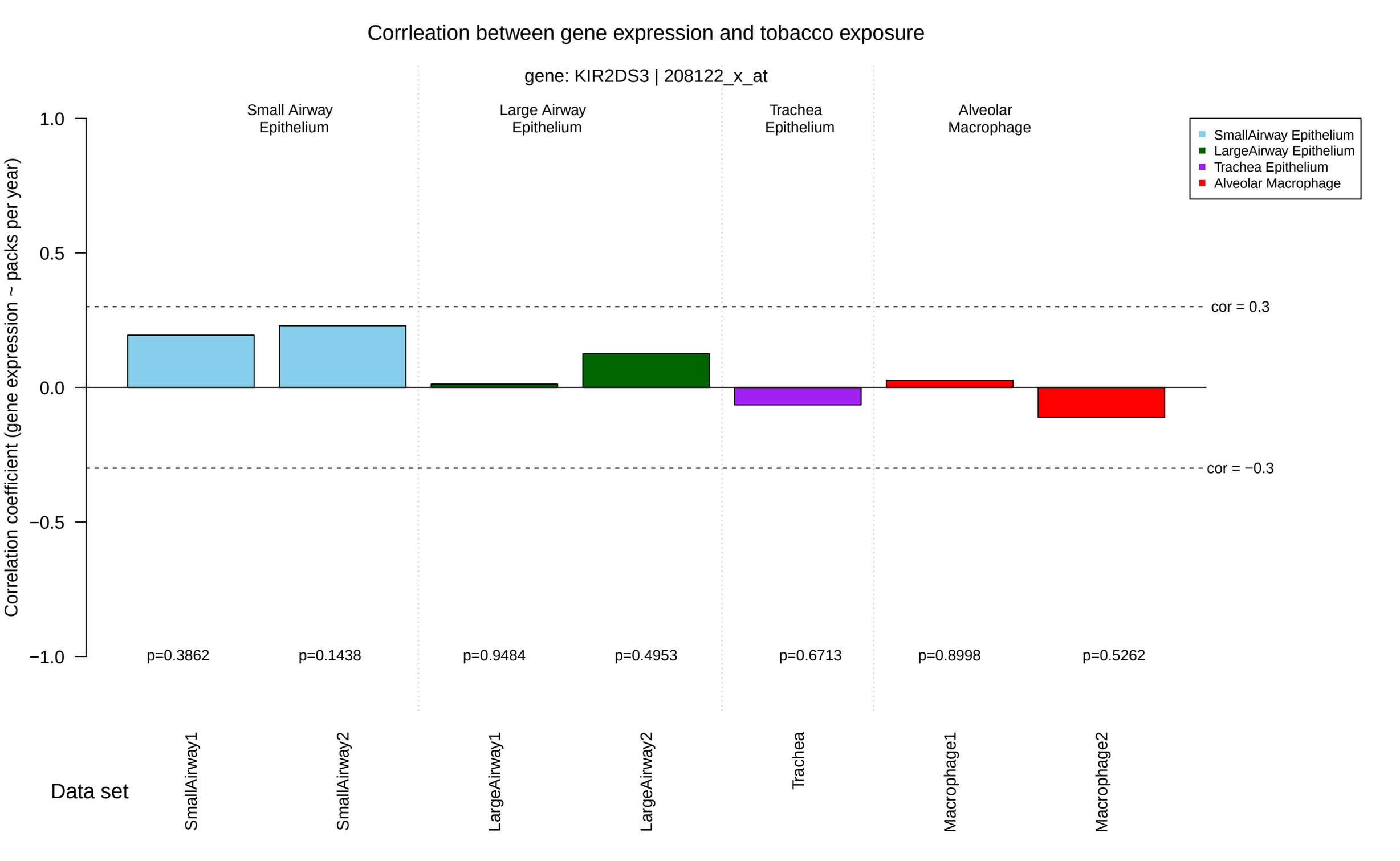

set: LargeAirway1 (p=0.5673)
KIR2DS3 | 208122_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3088)
KIR2DS3 | 208122_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6266)
KIR2DS3 | 208122_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9769)
KIR2DS3 | 208122_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4545)
KIR2DS3 | 208122_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIR2DS3 | 208122_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3862
p=0.1438
p=0.9484
p=0.4953
p=0.6713
p=0.8998
p=0.5262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2